

X-Ray Experiment

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Abstract—Using an X-ray tube with a molybdenum target, we investigated both the continuous Bremsstrahlung spectrum and the characteristic K_α and K_β lines. Bragg diffraction with a NaCl crystal enabled wavelength determination, while calibration of angular misalignment improved accuracy. Linear fits to the Bremsstrahlung edge yielded cutoff wavelengths consistent with the Duane–Hunt law. Converting these values to frequencies and plotting $E = eV$ versus $\nu_{\max}$ allowed an experimental extraction of Planck’s constant. Although uncertainties in voltage and angle introduced large statistical errors, the resulting value remained of the correct order of magnitude.

I. INTRODUCTION

This experiment combines too many different works on radiation. It includes X-ray radiation, the Bremsstrahlung effect, the Duane–Hunt Law, Bragg’s Law and Kramer’s Rule, which we will describe first their historical development and then theoretical connections.

HISTORY

X-rays were first found by Wilhelm Conrad Röntgen in 1895 when he was working with Crookes tubes[1]. He noticed that the fluorescent screen was shining during his experiments. He then named those unknown radiations as X-Strahlen, which means X-radiation in German. Both the nature and the spectral behavior of X-rays were not fully understood in that time.

Charles Barkla was the first to understand the spectral properties of the X-rays. He showed[2] that the X-rays have characteristic lines (or namely peaks) depending on the element. That is, the spectrum was not only continuous but also has sharp peaks due to the shell transitions. Barkla found the K_α and K_β radiations. His work was foundational for the X-ray spectroscopy. In 1917, he was awarded the Nobel Prize in Physics “for his discovery of the characteristic Röntgen(X-ray) radiation of the elements”.

In 1913, Moseley found[3] that the frequency of the characteristic X-ray lines (K- and L-lines) is proportional to the atomic number. He described the relation as $\nu \propto (Z - 1)^2$. The atomic number was first properly defined, and his work led to the rearrangement of the atomic table.

In 1912–1913, W.H. Bragg and W.L. Bragg discovered[4] that the X-rays interfere on the surface of the crystals. This interference pattern could be projected onto a surface. They stated that the interference strongly depends on the incoming angle. After their work, characteristic X-ray wavelengths could be known via a rather simple angle measurement. In 1915, they were awarded the Nobel Prize in Physics “for their services in the analysis of crystal structure by means of X-rays”.

In 1915, William Duane and Franklin Hunt found[5] that the X-ray spectrum has its own lowest possible wavelength. That

is, the spectrum for X-ray radiation has a cut-off wavelength, and they stated that this wavelength depends on the voltage applied to the tube. They also provided an independent way to obtain (or to test) the value of Planck’s constant, which we also will use in this experiment.

The continuous spectrum of X-ray radiation has been seen to be a bit odd since it was first found. It was started to be called “the Bremsstrahlung effect” around 1900s, which is translated to English as “the braking radiation effect”. In 1923, the first mathematically complete Bremsstrahlung[6] theory was suggested by Hendrik A. Kramers[7]. His theory also demonstrates that the Bremsstrahlung spectrum shows a linear behavior near the cut-off wavelength that was found by the Duane–Hunt Law.

THEORY

We talked about the physical law and effect and also their history briefly in the previous section. In this section, we try to give a comprehensive theoretical knowledge that will be tested in the experiment.

We first explain what is going on in the X-ray tube. The electrons generated by an electron gun are accelerated with an applied voltage V_{acc} through the element that is responsible for the creation of the X-rays, inside the tube. The total energy of each electron is given by the following equation.

$$E = eV_{\text{acc}} \quad (1)$$

where e and V_{acc} are the electron charge and accelerating voltage, respectively.

When an electron interacts with the nucleus of the atoms of the anode material, two different types of radiation occur, one of which is the Bremsstrahlung and the other one is characteristic radiation. In the following subsections, we explain both types of radiation.

The Bremsstrahlung

We know that the accelerating charged particles radiate energy. This radiation was the foundation of the classical electromagnetic theory. However, it was not enough to explain the radiation generated by the electron because due to its speed, the relativistic corrections gain a huge importance. Thus, it is given by the relativistic Liénard formula, which is

$$P(t) = \frac{2e^2\gamma^4}{3c} \left(\dot{\beta}^2 + \frac{(\boldsymbol{\beta} \cdot \dot{\boldsymbol{\beta}})^2}{1 - \beta^2} \right) \quad (2)$$

where $\boldsymbol{\beta} = \mathbf{v}/c$ and $\gamma = 1/\sqrt{1 - \beta^2}$.

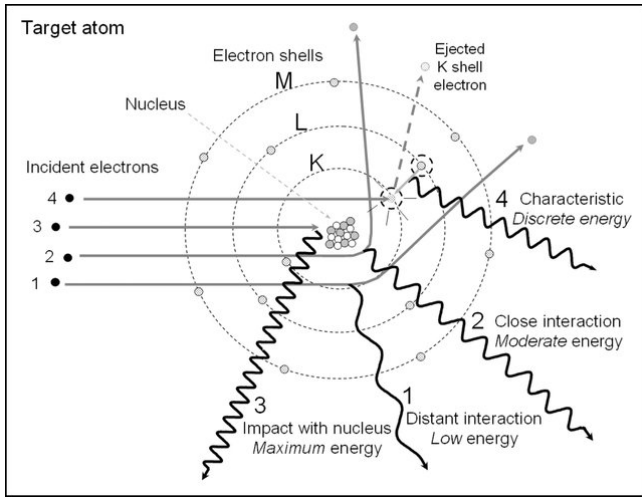


Fig. 1. This figure illustrates the different possible scenarios in the X-ray production process. 1, 2 and 3 shows how the Bremsstrahlung spectrum is generated. The 4th one shows the emission of the characteristic radiation.

This formula gives us the total radiation power of an electron under the influence of an accelerating field. However, it is not possible to tell the spectral distribution of the radiated energy. To obtain information on the spectrum, we need to analyze Eq.2 in the frequency domain. It is not straightforward and thus pretty complicated analysis, so we directly skip to Kramer's formula. The shape of the continuous X-ray spectrum is "approximately" given by Kramer's formula[7], however, this approximation is pretty valid.

$$I(\lambda) d\lambda = Z \left(\frac{\lambda}{\lambda_{\min}} - 1 \right) \frac{1}{\lambda^2} d\lambda \quad (3)$$

where the value of $\lambda_{\min}$ is given by the Duane-Hunt Law, which simply states that the configuration for a photon to gain the maximum possible energy via Bremsstrahlung is 3rd in Fig.1. That is, if an electron spends its whole energy to create an X-ray, the wavelength of the created X-ray is given by[5]

$$\lambda_{\min} = \frac{c}{\nu_{\max}}, \quad (4)$$

$$h\nu_{\max} = eV_{\text{acc}}, \quad (5)$$

$$\Rightarrow \lambda_{\min} = \frac{hc}{eV_{\text{acc}}}. \quad (6)$$

Thus, the cut-off frequency is highly dependent on the voltage supplied to the tube.

In Fig.1, we can see that the energy of the emitted photon during the Bremsstrahlung process depends on the impact parameter of the incoming electron. The smaller the impact parameter, the higher the energy that a photon can have. Therefore, if we think of the radiation process in terms of the acceleration of the electron, since the acceleration is a continuous function, then it is natural to think that the spectrum of the Bremsstrahlung is also continuous. On the other hand, this is not the only process seen in the same figure. The other is the characteristic radiation.

The Characteristic Radiation

In some instance, as seen in Fig.1, an electron may knock off the shell electron. Therefore, this vacancy should be filled by any electron that sits on the higher shell, since the lower shell has less energy and is more stable. This transition emits a photon, namely the characteristic x-ray. The frequency of the radiated photon is given by the following equation.

$$h\nu = E_{\text{initial}} - E_{\text{final}} \quad (7)$$

Since the electrons in the specific shell have a specific energy, and since the energies of the shells are discrete, the characteristic radiation can only attain particular wavelengths. In this experiment, we only observe the K_{α} and the K_{β} lines. Why do we not observe the higher transitions? The answer is the transition probability.

$$f_{n \rightarrow 1} \sim \frac{1}{n^3} \quad (8)$$

The K_{α} transition is from $L \rightarrow K$ and the K_{β} transition is from $M \rightarrow K$, where the shell numbers of L and M are $n = 2$ and $n = 3$, respectively. In Eq.8, it can be seen that for $n \geq 4$, the transition probability decreases significantly. Thus, we only observe the first two lines.

Now, we know how the X-rays are created in the tube. In the previous two sections, we examined their creation processes. The next question is how we measure both intensity and wavelength. As discussed previously in the history part, by use of Bragg's Law, we transform the whole process into a simple angle measurement experiment.

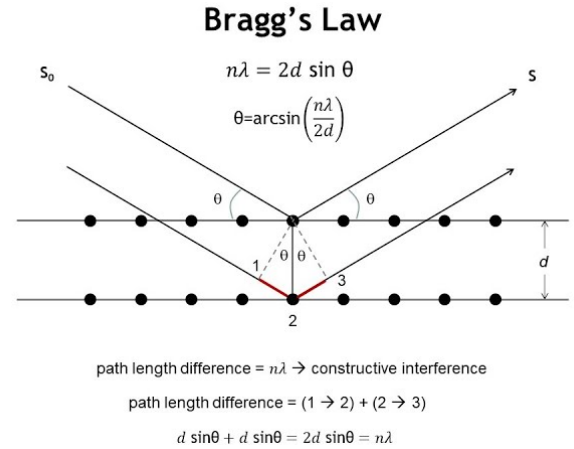


Fig. 2. This figure shows the Bragg diffraction.

Since the Bremsstrahlung has a continuous wavelength spectrum, in Bragg's Law, it also has a continuous angle spectrum. Similarly, since the characteristic X-ray emissions are discrete, in Bragg's Law, we should see a set of angles where the intensity has a sharp peak. This is because for different $n = 1, 2, \dots$ values, characteristic peaks appear more than once in the Bragg spectrum. It is given by[4]

$$n\lambda = 2d \sin \theta \quad (9)$$

where n is an integer and d is the structure distance of the crystal. Using all these, we can deduce the Planck's constant (h) as predicted by the Duane-Hunt Law.

II. METHOD

In this section, we give the step-by-step conducting procedure of the experiment.[8, 9]

In the experiment, we just give the initial conditions from the computer, then the setup does everything by itself. Here, you can find what the initial conditions should be selected and why we choose them.

- 1) First, run both the computer and the experiment setup. Then open the app designed for the experiment on the computer.
- 2) In the first run, we collect the data to calibrate our angle measurements. To do this, run the setup with the following initial conditions.
 - **Type of measurement:** Spectra
 - **X-data:** Crystal angle
 - **Emission current:** 0.8mA
 - **Integration time:** 5s
 - **Voltage:** Constant voltage at 35kV
 - **Rotation mode:** 2:1 coupled mode (which allows the Geiger-Müller Tube to rotate at twice the angle that the crystal rotates)
 - **Crystal angle:** 3° starting angle; 45° stopping angle (when the crystal is at 45°, the Geiger-Müller Tube is at 90°); 0.1° angle increment
 - **Setup:** Crystal is NaCl (100); no absorber; no filter
 - **Displays:** Select all
- 3) Run the device with the initial conditions above. Collect the first data set. Apply a Gaussian fit to each characteristic peak of the X-Ray source material (which is Molybdenum for this experiment). Calculate the corresponding angles using the known wavelength values for the characteristic radiations. Compare the angle values that you obtain both experimentally and theoretically, and construct a calibration function. This function will be helpful for the remaining part of the experiment.
- 4) Now, we collect 6 data sets to observe the Duane-Hunt Displacement Law. Set the stopping angle much smaller (for example 20° should be enough) for the rest of the experiment.
- 5) Run the device 6 times. Each time, change the constant voltage value starting from 15kV with 3kV steps, ending at 30kV, and save the data set at the end of each run. We will use these 6 data sets at the rest of the experiment. You can discard the data set you obtain to calibrate the angle.
- 6) Calculate the wavelength values using Bragg's Law and the calibration function you constructed in part 3. Plot the wavelength values and apply a linear fit at the linearly increasing part of your graph. Extract the value $\lambda_{\min}$, which corresponds to the intersection of the fit line with the x-axis. Repeat the process for each of your data sets.

- 7) Calculate $E = qV_{\text{acc}}$ and convert $\lambda_{\min}$ to $\nu_{\max}$ for each of your data sets.
- 8) Plot E vs. $\nu_{\max}$ graph, and the slope of this graph should give you the value of Planck's constant h .

III. THE EXPERIMENTAL SETUP

In this section, we will give the names and functions of all the apparatuses that are used in the experimental setup[8, 9]. Note that the setup we used in the experiment is very automated, so most of the work is done by the computer automatically.

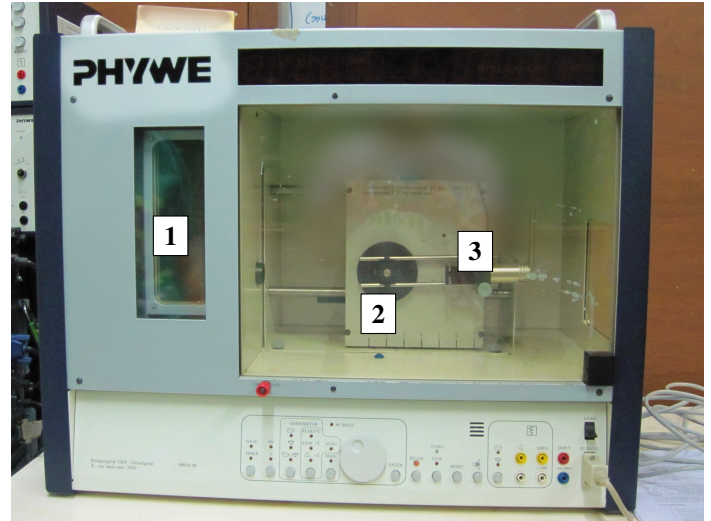


Fig. 3. X-Ray Experimental Setup

- 1) **X-Ray Creation Unit:** This chamber is used to create X-Rays. More detailed information on how it works and what the interior of the chamber looks like is found in Fig.4 below.
- 2) **Diffraction Crystal (NaCl) and Rotation Frame:** This crystal is used to diffract the incoming X-Rays. The crystal is placed onto a platform connected to a shaft that is able to rotate around its axis. As it rotates at an angle θ around its axis, the Geiger-Müller Tube, which is shown with number 3 in Fig.3 placed at the end of the shaft, rotates at an angle 2θ . This arrangement is necessary due to Bragg's Law. When a ray strikes the surface of a crystal at an angle θ , it diffracts at an angle 2θ .
- 3) **Geiger-Müller Tube:** It is used to detect the intensity of the diffracted waves. Depending on the wavelength (λ) of the incoming ray and the lattice distance (d) of the crystal, it reads very high intensities at some critical angles. More detailed information about the process is shared both in the theory and analysis parts. Furthermore, in Fig.5, the interior scheme and the working principle of the Geiger-Müller Tube is shown.
- 4) **Compute:** In this experiment, the computer is used to complete various works. The determination of angle ranges and initial voltages is done by the computer. It is also used to decode the digital data sets from Geiger-Müller Tube, which will later be used to create a graph.

In the figure below, we can see the interior of an X-Ray creation unit.

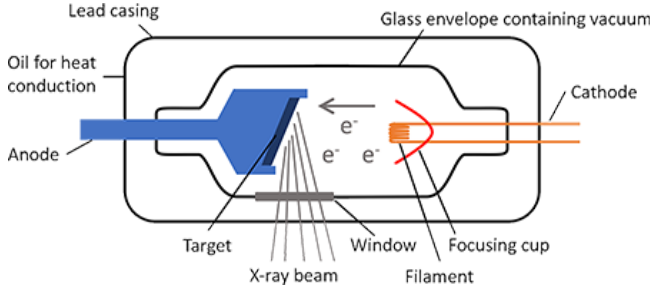


Fig. 4. The schematic depiction of an X-Ray creation tube.

Here, the cathode end is used to heat the filament. Then, the heated filament supplies free electrons to the evacuated chamber. These free electrons are focused in a very narrow direction. (This structure is very similar to an electron gun. In most of the X-Ray units, the electron gun is directly used.)

The free electrons that we obtained from cathode (or electron gun) are then accelerated through the anode end, where the target is placed at. The target element we used in this experiment is Molybdenum (Mo-42). From the interaction of electrons and the target element, X-Rays are formed. (Either via the Bremsstrahlung effect or de-excitation, more information is posted in the theory section.)

The newly created X-Rays are targeted into a very narrow slit. After it is successfully done, we have a very precise X-Ray beam with continuous broad spectrum with a well-defined cutoff.

In the following figure, we can see the scheme of the Geiger-Müller Tube. We will explain how it works.

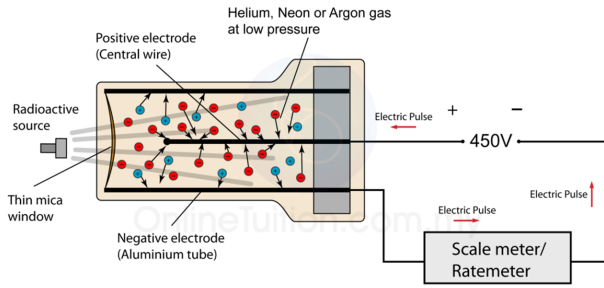


Fig. 5. The schematic depiction of an Geiger-Müller Tube.

In the left of the figure, it is shown that there are a radioactive source and the mica window. The radioactive source in our case is the emitted X-Rays. The mica window allows the incoming waves to pass through the window, but restricts the particles inside to pass to the outside of the window.

There is a noble gas mixture with low pressure inside the tube. The incoming wave, which has passed through the window, interacts with the gas atom and starts the ionization

process. The ionized particles are split into negatively and positively charged particles. Since the potential difference between the anode and the cathode ends is very high (approximately 450V), negatively charged particles accelerate at a very high rate towards the anode end, whereas positively charged particles drift towards the cathode end. This rapid acceleration of the particles leads to the ionization of the other atoms in the gas mixture, which will then be responsible for the avalanche effect. At the very end of this whole business, a strong impulse is created and the scale-meter reads a count. Please note that the Geiger-Müller Tube does nothing with the intensity of the impulse, it just counts them.

IV. THE DATA

In this section, we give our data sets. Since the data sets include too many data points, we share a google drive link to access the data sets rather than post them here as a table.

To access the drive click [here](#).

V. THE ANALYSIS

In this section, we perform the analysis of the experiment in detail.

In the first part of the analysis, we plot the data set taken at $V_{acc} = 35kV$ with error bars (see Fig.6). Then, since we know that Molybdenum has characteristic peaks (see Tab.I) at some critical angle values, we apply Gaussian fits onto these peaks in the plot. We will note the resulting the mean (μ) and standard deviation (σ) values for each peak. It is important to note that there are uncertainties for both μ and σ values, which are $\Delta\mu$ and $\Delta\sigma$, respectively. However, we will discard those statistical uncertainties. We know that theoretically, the characteristic peaks should appear as vertical lines. However, this is not the case because of instrumental uncertainties and background noises. Thus, for the peaks on the plot, a distributional width appears. We use these widths (known as std. dev. of the Gaussian fit (σ)) as uncertainties of the peaks.

TABLE I
CHARACTERISTIC X-RAY EMISSION LINES FOR MOLYBDENUM[10]

	K_{α_1}	K_{α_2}	K_{α}	K_{β}
E (keV)	17.479	17.374	17.444	19.607
λ (nm)	0.070940	0.071358	0.071062	0.063248

The K_{α} lines are the consequence of the transitions of the electrons from the $L(n = 2)$ shells to the $K(n = 1)$ shell. There are 3 different L shells, which are L_1 , L_2 , L_3 . Here, we are interested in the transitions specifically from the L_3 and L_2 shells, which are called the K_{α_1} and K_{α_2} lines. As it can be seen in Tab.I above, the energy values (that is, the wavelengths) of these two lines are very close to each other.

Furthermore, the ratio of the transition intensities for these two lines is almost 2:1 for all elements (regarding non-relativistically). The transition intensity is proportionally given by

$$I_{i \rightarrow f} \propto g_i A_{i \rightarrow f} \quad (10)$$

where g_i is the degeneracy of the initial orbit and $A_{i \rightarrow f}$ is the transition amplitude.

The degeneracy is given by $g = 2j + 1$, where j is the total angular momentum quantum number.

$$L_2 : \quad j = \frac{1}{2} \rightarrow g_{L_2} = 2 \quad (11)$$

$$L_3 : \quad j = \frac{3}{2} \rightarrow g_{L_3} = 4 \quad (12)$$

Since the transition amplitudes from L_3 and L_2 are almost the same (i.e. $A_{L_3 \rightarrow K} \approx A_{L_2 \rightarrow K}$), then the ratio of the intensities becomes

$$\frac{I_{K_{\alpha_1}}}{I_{K_{\alpha_2}}} \equiv \frac{I_{L_3 \rightarrow K}}{I_{L_2 \rightarrow K}} \approx \frac{g_{L_3}}{g_{L_2}} = \frac{4}{2} = 2. \quad (13)$$

We cannot distinguish the K_{α_1} and K_{α_2} lines in the plot. So, we take their average with considering their weights in the contribution to the total intensity. Then K_{α} is given by the following equation.

$$K_{\alpha} := \frac{2}{3}K_{\alpha_1} + \frac{1}{3}K_{\alpha_2} \quad (14)$$

We know that the beams that are diffracted from the surface of a crystal satisfy Bragg's Law. Thus, the characteristic X-Ray beams should also satisfy it, and we hope to be able to observe the constructive interference of these lines at some angles. Bragg's Law[4] is given by

$$n\lambda = 2d \sin \theta. \quad (15)$$

In the following Tab.II, you can see the first 3 angles for both K_{α} and K_{β} lines, at which the constructive interference occurs.

TABLE II
CORRESPONDING ANGLES FOR K_{β} AND K_{α} AT DIFFERENT n VALUES

	K_{β}	K_{α}
$\theta (n = 1)$	6.4388°	7.2383°
$\theta (n = 2)$	12.9608°	14.5955°
$\theta (n = 3)$	19.6593°	22.2092°

In Fig.6 below, we plot the data set with error bars and apply Gaussian fits to the characteristic peaks. We note the resulting mean and std. dev. values in Tab.III. We know that both the mean (μ) and the standard deviation (σ) have their own uncertainty values due to statistical errors. Therefore, as we mentioned earlier, the characteristic peaks should appear as vertical lines. However, in reality, this is not the case. So, here we decided to take the σ values as the uncertainty of the corresponding peaks regardless of their own uncertainties.

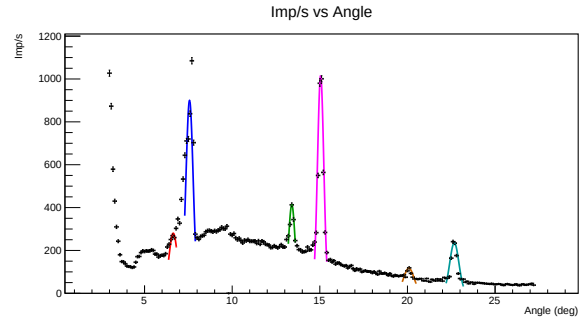


Fig. 6. The figure shows the Imp/s vs. angle graph for the data set collected at $V_{acc} = 35\text{kV}$. We also apply 6 different Gaussian fits onto the peaks on the graph. Those peaks are the characteristic peaks of Molybdenum. For the mean and the standard deviation for each Gaussian fit, see Tab.III below.

TABLE III
RESULTING VALUES OF FITS FOR EACH PEAK.

Fits	Mean (μ)	Std. Dev. (σ)	χ^2/NDF
1 st (red)	6.63739	0.245279	1.29466
2 nd (blue)	7.56757	0.203467	1.453
3 rd (green)	13.4023	0.185854	0.110919
4 th (magenta)	15.0465	0.178067	0.206203
5 th (brown)	20.0792	0.300065	1.1076
6 th (cyan)	22.6736	0.260826	0.930257

Now, using the values in Tab.II and Tab.III, we construct a calibration function. This procedure is necessary because the calibration of the experimental instruments may be deformed due to frequent use or they may have intrinsic off-sets. To minimize the effects of the instrumental uncertainties of the experimental setup, we continue the rest of the analysis using the calibration function we constructed below.

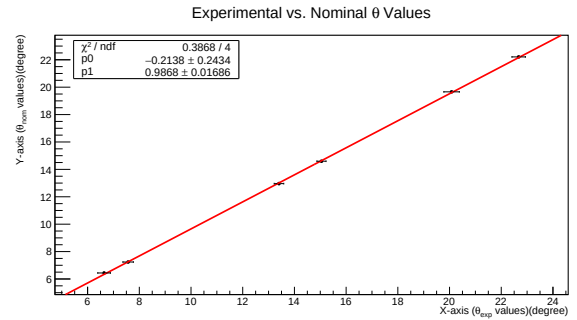


Fig. 7. The figure illustrates the correspondence between nominal and experimental angle values. The calibration function is constructed using the slope and the interception data of the linear fit. Note that the data points have the error bars only along the x-axis. This is because we interpret the nominal values as true values.

The calibration function is in the following form.

$$\theta_{\text{calibrated}} = p_1 \theta_{\text{experimental}} + p_0 \quad (16)$$

where $p_1 = 0.9868 \pm 0.01686$ and $p_0 = -0.2138 \pm 0.2434$.

The other thing we want to discuss is the χ^2/ndf test. Here, in this fit, you can see that $\chi^2/\text{ndf} = 0.3868/4 = 0.0967$,

which is a very small value for a fit to be reliable. In the fit, the experimental values and nominal values form almost a perfect line. However, in this setup, the errors of the experimental values are so big that the algorithm thinks the fit is perfect, which is called the overfit. Thus, the fit we made suffers from overfitting. For a fit to be reliable, $\chi^2/\text{ndf} \approx 1$.

This is the second part of the analysis. In this part of the analysis, we apply a linear fit to the linear region of the plots with different V_{acc} to obtain λ_{min} values. We test the Duane-Hunt Displacement Law with the help of Kramer's Rule. First, we transform all angle (θ) data to wavelength (λ) data by Eq.15. We also use the calibration function we found in Eq.16 to calibrate our angle data points. Note that p_1 and p_0 also have their own uncertainties, so the uncertainty in the calibrated angle is given by

$$\sigma_{\theta_{\text{cal}}}^2 = \sigma_{p_1}^2 \theta_{\text{exp}}^2 + \sigma_{\theta_{\text{exp}}}^2 p_1^2 + \sigma_{p_0}^2. \quad (17)$$

Using this, the error propagation for the wavelength (λ) in Eq.15 is given by

$$\sigma_{\lambda}^2 = \sigma_{\theta_{\text{cal}}}^2 \left(\frac{2d}{n} \cos \theta_{\text{cal}} \right)^2 \quad (18)$$

We again use the same error for Imp/s data. Imp/s data are taken in 5 second intervals. Thus, the number of total events is actually the 5 times of the Imp/s. Since the error in counting experiments is given by $\sigma = \sqrt{\mu}$, in this case, it is

$$\sigma_{\text{Imp/s}} = \sqrt{\# \text{ of total event}} = \frac{\sqrt{5 \text{ Imp/s}}}{5} = \sqrt{\frac{\text{Imp/s}}{5}} \quad (19)$$

After plotting data sets with errors calculated above, we apply a linear fit to the linear region in each plot. This is because the Kramer's Law suggests that the intensity of the emitted photon in the Bremsstrahlung[7] process is given by

$$I(\lambda) = Z \left(\frac{\lambda}{\lambda_{\text{min}}} - 1 \right) \frac{1}{\lambda^2}, \quad (20)$$

where Z is the atomic number and λ_{min} is the minimum wavelength predicted by the Duane-Hunt Law, which states

$$\nu_{\text{max}} = \frac{eV_{\text{acc}}}{h} \implies \lambda_{\text{min}} = \frac{hc}{eV_{\text{acc}}}. \quad (21)$$

Eq.20 shows a linear behavior near the value λ_{min} .

$$I(\lambda) \approx \frac{Z}{\lambda_{\text{min}}^3} (\lambda - \lambda_{\text{min}}) + 2 \frac{Z}{\lambda_{\text{min}}^4} (\lambda - \lambda_{\text{min}})^2 + \dots \quad (22)$$

Thus, we extract the intersection point of the line with the x-axis and declare it as λ_{min} .

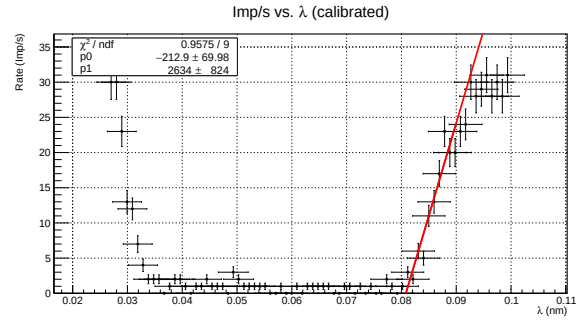


Fig. 8. The figure illustrates the relation between Imp/s and wavelength at $V_{\text{acc}} = 15\text{kV}$. A linear fit is applied to the linear region of the Bremsstrahlung.

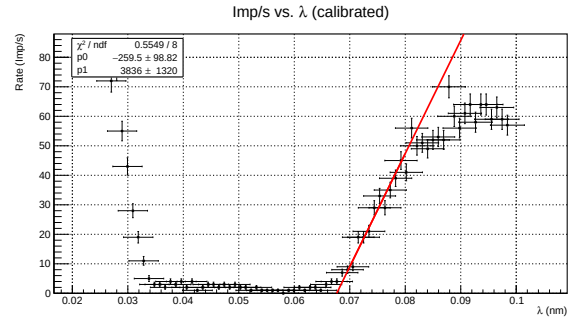


Fig. 9. The figure illustrates the relation between Imp/s and wavelength at $V_{\text{acc}} = 18\text{kV}$. A linear fit is applied to the linear region of the Bremsstrahlung.

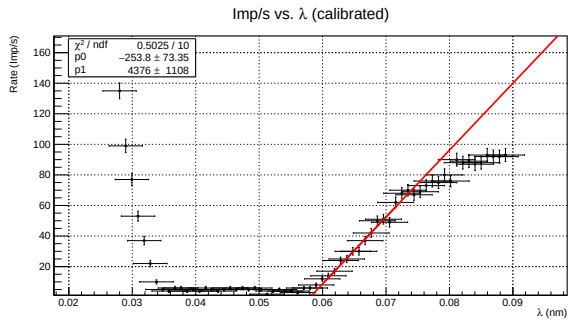


Fig. 10. The figure illustrates the relation between Imp/s and wavelength at $V_{\text{acc}} = 21\text{kV}$. A linear fit is applied to the linear region of the Bremsstrahlung.

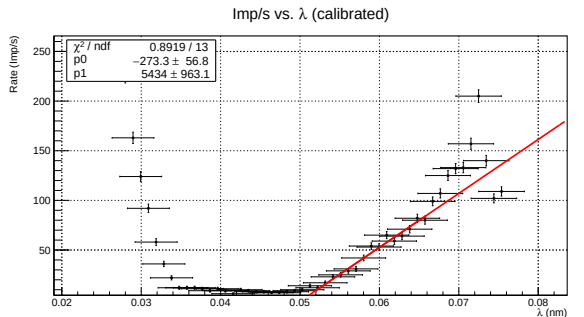


Fig. 11. The figure illustrates the relation between Imp/s and wavelength at $V_{\text{acc}} = 24\text{kV}$. A linear fit is applied to the linear region of the Bremsstrahlung.

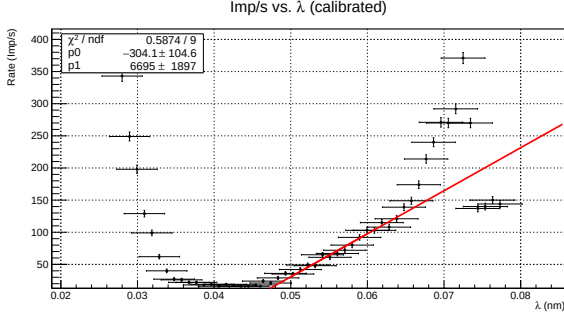


Fig. 12. The figure illustrates the relation between Imp/s and wavelength at $V_{acc} = 27\text{kV}$. A linear fit is applied to the linear region of the Bremsstrahlung.

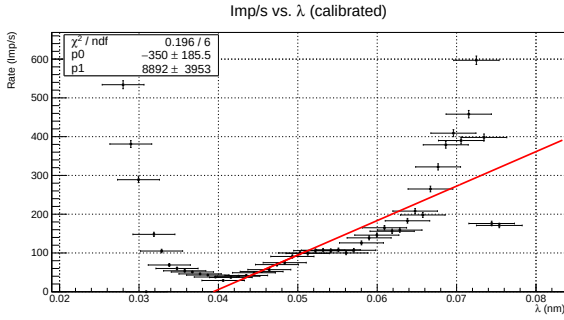


Fig. 13. The figure illustrates the relation between Imp/s and wavelength at $V_{acc} = 30\text{kV}$. A linear fit is applied to the linear region of the Bremsstrahlung.

For all plots, we take $n = 1$ in the calculation of the wavelength via Bragg's Law. As seen in the data sets, for characteristic peaks, the $n > 1$ peaks are not in the range. Moreover, since the Bremsstrahlung effect provides a continuous spectrum, there always is a corresponding wavelength at all frequencies at every angle. However, the intensity of the Bremsstrahlung for $n \geq 2$ is significantly lower than for $n = 1$. These are the reasons why we take $n = 1$ in the calculation.

Furthermore, the linear fit function is in the following form.

$$\text{Imp/s} = p_1 \lambda + p_0 \quad (23)$$

$$0 = p_1 \lambda_{\min} + p_0 \quad (24)$$

$$\lambda_{\min} = \frac{-p_0}{p_1} \quad (25)$$

The error is given by

$$\sigma_{\lambda_{\min}}^2 = \frac{\sigma_{p_0}^2}{p_1^2} + \frac{\sigma_{p_1}^2 p_0^2}{p_1^4}. \quad (26)$$

In the following table, we list the $\lambda_{\min}$ values obtained from 6 different plots.

TABLE IV
RESULTS FOR EACH GRAPH

Voltage	p_1	p_0	χ^2/ndf	$\lambda_{\min}$ (nm)
15kV	2634 ± 824	-212.9 ± 69.98	0.9575/9	0.08081 ± 0.03667
18kV	3836 ± 1320	-259.5 ± 98.82	0.5549/8	0.06765 ± 0.03472
21kV	4376 ± 1108	-253.8 ± 73.35	0.5025/10	0.05801 ± 0.02229
24kV	5434 ± 963.1	-273.3 ± 56.8	0.8919/13	0.05031 ± 0.01374
27kV	6695 ± 1897	-304.1 ± 104.6	0.5874/9	0.04542 ± 0.02024
30kV	8892 ± 3953	-350 ± 185.5	0.196/6	0.03936 ± 0.02723

Now, we are at the final part of the analysis. In this part, first we convert the $\lambda_{\min}$ values to $\nu_{\max}$ according to Eq.21. In Tab.V below, you can find the converted data. Then, we make the plot of the $E = eV_{acc}$ vs. $\nu_{\max}$ graph, and applying the linear fit, we will try to calculate the experimental value of Planck's constant. We take the uncertainty of the applied voltage as $\pm 1\text{kV}$.

TABLE V
CONVERTED FREQUENCY ($\nu_{\max}$) AND ENERGY ($E = eV_{acc}$) QUANTITIES WITH PROPER SIGNIFICANT FIGURES

Voltage (kV)	E (J)	$\nu_{\max}$ (Hz)
15	$(2.40 \pm 0.16) \times 10^{-15}$	$(3.71 \pm 1.69) \times 10^{18}$
18	$(2.88 \pm 0.16) \times 10^{-15}$	$(4.43 \pm 2.27) \times 10^{18}$
21	$(3.36 \pm 0.16) \times 10^{-15}$	$(5.17 \pm 1.99) \times 10^{18}$
24	$(3.85 \pm 0.16) \times 10^{-15}$	$(5.96 \pm 1.63) \times 10^{18}$
27	$(4.33 \pm 0.16) \times 10^{-15}$	$(6.60 \pm 2.94) \times 10^{18}$
30	$(4.81 \pm 0.16) \times 10^{-15}$	$(7.62 \pm 5.27) \times 10^{18}$

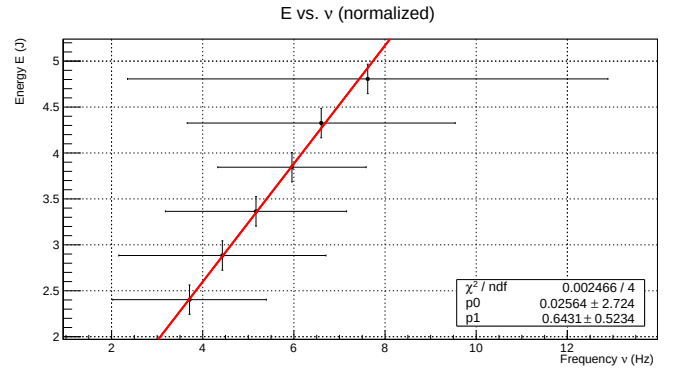


Fig. 14. This figure illustrates relation between the energy (E) and the maximum frequency ($\nu_{\max}$). Slope of the linear fit is equal to Planck's constant. Note that, here we normalize the axis for the line fit algorithm to work well. To convert the result to original, we should multiply p_1 and p_0 by 10^{-33} and 10^{-15} , respectively.

VI. THE RESULT

In this section, we share the results of the experiment. In the last part of the analysis, we applied the linear fit to the energy vs. frequency graph, which is obtained according to the Duane-Hunt Law, described in Eq.21. The slope of the line should give us the experimental value of Planck's constant. Note that we applied the line fit to a normalized graph, which means that we need to multiply p_0 and p_1 by 10^{-15} and 10^{-33} , respectively. In the following, you can find the experimental value we found.

$$h_{\text{exp.}} = (6.4 \pm 5.2) \times 10^{-34} \text{ J} \cdot \text{s} \quad (27)$$

The one important thing we should mention is that the results of the χ^2/ndf test were always very low throughout the entire analysis (except in the first part where we apply Gaussian fits). This is because the uncertainties of our data points were too high to make a reliable line fit. This causes a very high uncertainty in the final result.

The reliable experimental value of the Planck's constant in the literature is given below.

$$h_{\text{nominal}} = 6.62606957(29) \times 10^{-34} \text{ J} \cdot \text{s} \quad (28)$$

Our result is $\sim 0.042\sigma$ away from the nominal value. This is not because we deduce the experimental value so precise, instead, this is because our uncertainty is very big. However, we found its value at the correct order, that is, $h \sim 6 \times 10^{-34} \text{ J} \cdot \text{s}$.

On the other hand, we need to enhance the experimental setup so that the uncertainty of the result decreases. To accomplish this, firstly, we can calibrate the setup by hand to avoid possible off-sets in the angle measurement. One further thing on this is to take the measurements in smaller intervals, i.e. we should take the data with 0.05 increment in angle (it would be better if it were smaller). On the other hand, we have a very high uncertainty in the accelerating voltage values. This may be fine tuned.

Moreover, the experiment also suffers from the statistical uncertainties. We needed to propagate errors for too many steps with an increasing number of sources of error at each step. We may try to take an average at some point in the analysis to discard these kinds of uncertainty. The other possible way is to find another approach. That is, when converting the θ values to the λ values, we are left with significant errors. This procedure may be replaced by another more sophisticated approach.

Lastly, we want to discuss the meaning of non-zero p_0 value. First, remember what the linear fit gives us.

$$E = h_{\text{eff}}\nu + E_0 \quad (29)$$

Theoretically correct formula is given by $E = h\nu$. This tells us that in Eq.29, E_0 , which is p_0 in linear fit, should be zero. However, this is not the case. This situation has two main reasons. One of which is the calibration off-set, which means that the X-Ray source or electron gun is not calibrated well. The other is the insufficiency of our model to explain the reality. There may be other background interactions (or processes) we did not include into our model, so that in the linear approximation, these effects are shown as ground level energy. In our case, it is mostly due to the lack of calibration and error propagation.

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VII. APPENDIX

```
#include <TCanvas.h>
#include <TGraphErrors.h>
#include <TFl.h>
#include <fstream>
#include <sstream>
#include <vector>
#include <iostream>
#include <cmath>

void calib_fits() {

    // read file
    std::ifstream
        infile("C:/Users/hp/Desktop/phys
        443/x-ray/Xray_35kv.txt");
    if (!infile.is_open()) {
        std::cerr << "File is not found" <<
            std::endl;
        return;
    }

    std::vector<double> angle, rate;
    std::string line;

    // skip headers
    std::getline(infile, line);
    std::getline(infile, line);

    while (std::getline(infile, line)) {

        std::stringstream ss(line);

        double th, r;
        if (ss >> th >> r) {
            angle.push_back(th);
```

```

        rate.push_back(r);
    }
}

int N = angle.size();

// errors
std::vector<double> errx(N, 0.1);
std::vector<double> erry(N);
for (int i = 0; i < N; i++) {
    erry[i] = std::sqrt(rate[i] / 5.0);
}

TCanvas* c1 = new TCanvas("c1", "Imp/s vs
    Angle", 1200, 800);

// graph with error
TGraphErrors* g = new TGraphErrors(N,
    angle.data(), rate.data(), errx.data(),
    erry.data());
g->SetTitle("Imp/s vs Angle;Angle
    (deg); Imp/s");
g->SetMarkerStyle(20);
g->SetMarkerSize(0.7);
g->Draw("AP");

// fit 1
TF1* fit1 = new TF1("fit1", "gaus", 6.37,
    6.82);
fit1->SetLineColor(kRed);
fit1->SetParameters(250, 6.8, 0.8);
g->Fit(fit1, "R+");

// fit 2
TF1* fit2 = new TF1("fit2", "gaus", 7.29,
    7.9);
fit2->SetLineColor(kBlue);
fit2->SetParameters(1150, 7.88, 0.5);
g->Fit(fit2, "R+");

// fit 3
TF1* fit3 = new TF1("fit3", "gaus", 13.2,
    13.6);
fit3->SetLineColor(kGreen + 2);
fit3->SetParameters(420, 13.3, 0.1);
g->Fit(fit3, "R+");

// fit 4
TF1* fit4 = new TF1("fit4", "gaus", 14.7,
    15.4);
fit4->SetLineColor(kMagenta);
fit4->SetParameters(1000, 15, 0.3);
g->Fit(fit4, "R+");

// fit 5
TF1* fit5 = new TF1("fit5", "gaus", 19.7,
    20.5);
fit5->SetLineColor(kOrange + 2);
fit5->SetParameters(150, 20, 0.2);
g->Fit(fit5, "R+");

// fit 6
TF1* fit6 = new TF1("fit6", "gaus", 22.2,
    23.2);
fit6->SetLineColor(kCyan + 2);
fit6->SetParameters(250, 22.5, 0.2);
g->Fit(fit6, "R+");

c1->Update();

```

```

// print fits
std::cout << "\nFit1 mean=" <<
    fit1->GetParameter(1)
    << ", sigma=" << fit1->GetParameter(2)
    << "\n";

std::cout << "Fit2 mean=" <<
    fit2->GetParameter(1)
    << ", sigma=" << fit2->GetParameter(2)
    << "\n";

std::cout << "Fit3 mean=" <<
    fit3->GetParameter(1)
    << ", sigma=" << fit3->GetParameter(2)
    << "\n";

std::cout << "Fit4 mean=" <<
    fit4->GetParameter(1)
    << ", sigma=" << fit4->GetParameter(2)
    << "\n";

std::cout << "Fit5 mean=" <<
    fit5->GetParameter(1)
    << ", sigma=" << fit5->GetParameter(2)
    << "\n";

std::cout << "Fit6 mean=" <<
    fit6->GetParameter(1)
    << ", sigma=" << fit6->GetParameter(2)
    << "\n";
}

void calib_plot_linear()
{
    // calib vals
    double p1 = 0.9868;
    double p0 = -0.2138;

    double dp1 = 0.01686;
    double dp0 = 0.2434;

    double dtheta_exp = 0.1;

    double d = 0.282; // Bragg
    int n = 1;

    // read file
    const char* filename =
        "C:/Users/hp/Desktop/phys_443/x-ray/Xray_30kv.txt";

    std::ifstream infile(filename);
    if (!infile.is_open()) {
        std::cerr << "File could not be opened."
            << std::endl;
        return;
    }

    std::vector<double> theta_exp_vec;
    std::vector<double> rate_vec;

    std::string line;

    // skip headers
    std::getline(infile, line);
    std::getline(infile, line);

    while (std::getline(infile, line)) {

```

```

    if (line.empty()) continue;

    std::stringstream ss(line);
    double theta_val, rate_val;

    if (!(ss >> theta_val >> rate_val))
        continue;

    theta_exp_vec.push_back(theta_val);
    rate_vec.push_back(rate_val);
}
infile.close();

int N = theta_exp_vec.size();
if (N == 0) {
    std::cerr << "File empty." << std::endl;
    return;
}

// theta calib
std::vector<double> theta_cal_vec(N);
std::vector<double> dtheta_cal_vec(N);

for (int i = 0; i < N; ++i) {
    double theta_exp = theta_exp_vec[i];

    double theta_cal = p1 * theta_exp + p0;
    theta_cal_vec[i] = theta_cal;

    double var_theta_cal =
        theta_exp * theta_exp * dp1 * dp1 +
        dp0 * dp0 +
        p1 * p1 * dtheta_exp * dtheta_exp;

    dtheta_cal_vec[i] =
        std::sqrt(var_theta_cal);
}

// lambda calc
std::vector<double> lambda_vec(N);
std::vector<double> dlambda_vec(N);

const double deg2rad = M_PI / 180.0;

for (int i = 0; i < N; ++i) {
    double theta_deg = theta_cal_vec[i];
    double dtheta_deg = dtheta_cal_vec[i];

    double theta_rad = theta_deg * deg2rad;

    double lambda = 2.0 * d *
        std::sin(theta_rad) / n;
    lambda_vec[i] = lambda;

    double dlambda_dtheta_deg =
        2.0 * d * std::cos(theta_rad) *
        deg2rad / n;

    dlambda_vec[i] =
        std::fabs(dlambda_dtheta_deg) *
        dtheta_deg;
}

// y errors
std::vector<double> y_err_vec(N);
for (int i = 0; i < N; ++i) {
    double y = rate_vec[i];
    if (y <= 0)
        y_err_vec[i] = 0;
    else
        y_err_vec[i] = std::sqrt(y / 5.0);
}

// graph
TGraphErrors* gr = new TGraphErrors(
    N,
    lambda_vec.data(),
    rate_vec.data(),
    dlambda_vec.data(),
    y_err_vec.data()
);

gr->SetTitle("Rate vs lambda");
gr->GetXaxis()->SetTitle("lambda (nm)");
gr->GetYaxis()->SetTitle("Rate (Imp/s)");

gr->SetMarkerStyle(20);
gr->SetMarkerSize(0.9);

TCanvas* c = new TCanvas("c", "Imp/s vs
    lambda", 900, 700);
c->SetGrid();
gr->Draw("AP");

// fit region
double fit_xmin = 0.043;
double fit_xmax = 0.051;

TF1* f = new TF1("f", "pol1", fit_xmin,
    fit_xmax);
gr->Fit(f, "RE");

// extend fit
double full_min = gr->GetXaxis()->GetXmin();
double full_max = gr->GetXaxis()->GetXmax();
f->SetRange(full_min, full_max);
f->SetLineColor(kRed);
f->Draw("same");

gStyle->SetOptFit(1);

// x-intercept
double b = f->GetParameter(0);
double m = f->GetParameter(1);
double db = f->GetParError(0);
double dm = f->GetParError(1);

double lambda_min = -b / m;

double dlambda_min =
    std::sqrt((db * db) / (m * m) + (b * b *
        dm * dm) / (m * m * m * m));

std::cout << "\nlambda_min = " << lambda_min
    << " +/- " << dlambda_min << " nm\n";

c->Update();
}

#include <TGraphErrors.h>
#include <TCanvas.h>
#include <TF1.h>
#include <TStyle.h>
#include <iostream>

void EvNu_fit()
{

```

```

gStyle->SetOptFit(1); // show fit box

// raw table
const int N = 6;

double E[N] = {
    2.40326495,
    2.88391794,
    3.36457093,
    3.84522392,
    4.32587691,
    4.80652990
};

double dE[N] = {
    0.1602176634,
    0.1602176634,
    0.1602176634,
    0.1602176634,
    0.1602176634,
    0.1602176634
};

double nu[N] = {
    3.711,
    4.431,
    5.169,
    5.958,
    6.600,
    7.620
};

double dnu[N] = {
    1.685,
    2.273,
    1.987,
    1.628,
    2.940,
    5.270
};

// graph
TGraphErrors* g = new TGraphErrors(N, nu,
    E, dnu, dE);
g->SetTitle("E vs nu ; nu (Hz) ; E (J)");
g->SetMarkerStyle(20);
g->SetMarkerSize(1.1);

// canvas
TCanvas* c = new TCanvas("c", "E vs nu",
    900, 700);
c->SetGrid();
g->Draw("AP");

// fit
TF1* f = new TF1("f", "pol1");
g->Fit(f, "E");

// extend fit
double xmin = g->GetXaxis()->GetXmin();
double xmax = g->GetXaxis()->GetXmax();
f->SetRange(xmin, xmax);
f->SetLineColor(kRed);
f->Draw("same");

double p0 = f->GetParameter(0);
double p1 = f->GetParameter(1);
double dp0 = f->GetParError(0);
double dp1 = f->GetParError(1);

```

```

std::cout << "\nfit results\n";
std::cout << "p0 = " << p0 << " +/- " <<
    dp0 << "\n";
std::cout << "p1 = " << p1 << " +/- " <<
    dp1 << "\n";

// physical units
double p0_phys = p0 * 1e-15;
double dp0_phys = dp0 * 1e-15;
double p1_phys = p1 * 1e-33;
double dp1_phys = dp1 * 1e-33;

std::cout << "\nphysical units\n";
std::cout << "p0_phys = " << p0_phys << "
    +/- " << dp0_phys << " J\n";
std::cout << "p1_phys = " << p1_phys << "
    +/- " << dp1_phys << " J/Hz\n";

c->Update();
}

```

```

#include <TGraphErrors.h>
#include <TF1.h>
#include <TCanvas.h>
#include <iostream>

void linefit_calibration()
{
    gStyle->SetOptFit(1);

    // x vals
    double x[6] = {
        6.63739,
        7.56757,
        13.4023,
        15.0465,
        20.0792,
        22.6736
    };

    // x errs
    double ex[6] = {
        0.245279,
        0.203467,
        0.185854,
        0.178067,
        0.300065,
        0.260826
    };

    // y vals
    double y[6] = {
        6.4388,
        7.2383,
        12.9608,
        14.5955,
        19.6593,
        22.2092
    };

    // y errs
    double ey[6] = {
        1e-6, 1e-6, 1e-6, 1e-6, 1e-6, 1e-6 };

    // graph
    TGraphErrors* g = new TGraphErrors(6, x, y,
        ex, ey);

```

```

g->SetTitle("theta exp vs theta nom;theta
exp (deg);theta nom (deg)");
g->SetMarkerStyle(20);
g->SetMarkerSize(1.1);

// canvas
TCanvas* c = new TCanvas("c", "Line Fit",
900, 700);
g->Draw("AP");

// fit
TF1* f = new TF1("f", "pol1");
g->Fit(f, "Q E");
}

```

The general formula for error propagation is given below.
Define a function

$$y = f(x_1, \dots, x_n). \quad (30)$$

Then the error σ_y of y is given as[8]

$$\sigma_y^2 = \sum_k^n \left(\frac{\partial f}{\partial x_k} \right)^2 \sigma_{x_k}^2 + \sum_{l < k}^m \left(\frac{\partial f}{\partial x_k} \right) \left(\frac{\partial f}{\partial x_l} \right) \sigma_{kl}^2 \quad (31)$$

where σ_{kl} represents the relative error of the variable x_k to x_l . Also, $\sigma_{kl} = \sigma_{lk}$.

If the variables do not depend on each other (relatively free or $\sigma_{kl} = 0$ for all k and l values), equation 31 becomes[8]

$$\sigma_y^2 = \sum_k^n \left(\frac{\partial f}{\partial x_k} \right)^2 \sigma_{x_k}^2. \quad (32)$$